Shichang Zhang

Contact Information

Address: 300 E Lemon St. BAC 522, Tempe, AZ 85287
Email: shichang.zhang@asu.edu
Website: https://shichangzh.github.io/

Work Experience

Arizona State University — Tempe, AZ
Assistant Professor of Information Systems, W.P. Carey School of Business — July 2026 - Present

Harvard University — Cambridge, MA
Postdoctoral Fellow, Harvard Business School AI Institute — Aug. 2024 - June 2026

Education

University of California, Los Angeles — Los Angeles, CA
Ph.D. in Computer Science — June 2024

Stanford University — Stanford, CA
M.S. in Statistics — Apr. 2019

University of California, Berkeley — Berkeley, CA
B.A. in Statistics — May 2017
Honors: Honors in Statistics, High Distinction

Research Interests

Explainable AI, Trustworthy AI, Data Attribution, Mechanistic Interpretability, Large Language Models, Graph Data Mining

Honors and Awards

NENLP **Outstanding Paper**	2025
KDD Outstanding Reviewer (Top 10%, twice, in Aug and Feb)	2025
Amazon PhD Fellowship	2023
J.P.Morgan Chase AI PhD Fellowship	2023
KDD Excellence in Reviewing (Top 30 of 1551)	2023
Snap Research Fellowship Honorable Mention	2022
ICML Top Reviewer (Top 10%)	2022
UCLA Graduate Division Fellowship	2021

Publications

Refereed Publications:

1. Who Gets Credit or Blame? Attributing Accountability in Modern AI Systems
Shichang Zhang, Hongzhe Du, Jiaqi W. Ma, Himabindu Lakkaraju
International Conference on Machine Learning (**ICML**), 2026

2. Explainability Research Must Prioritize Foundations over Ad-hoc Methods
Michal Moshkovitz, Suraj Srinivas, Lesia Semenova, Nave Frost, Cyrus Rashtchian, Valentyn Boreiko, **Shichang Zhang**, Himabindu Lakkaraju, Cynthia Rudin, Jennifer Wortman Vaughan
International Conference on Machine Learning (**ICML**), 2026

3. Efficient Ensembles Improve Training Data Attribution
   Junwei Deng*, Ting-Wei Li*, **Shichang Zhang**, Jiaqi Ma (*equal contribution)
   Transactions on Machine Learning Research (**TMLR**), 2026
4. A Survey on Graph Neural Network Acceleration: Algorithms, Systems, and Customized Hardware
   Shichang Zhang, Atefeh Sohrabizadeh, Cheng Wan, Zijie Huang, Ziniu Hu, Yewen Wang, Yingyan (Celine) Lin, Jason Cong, Yizhou Sun
   ACM Computing Survey (**CSUR**), 2026
5. How Post-Training Reshapes LLMs: A Mechanistic View on Knowledge, Truthfulness, Refusal, and Confidence
   Hongzhe Du*, Weikai Li*, Min Cai, Karim Saraipour, Zimin Zhang, Himabindu Lakkaraju, Yizhou Sun, **Shichang Zhang** (*equal contribution)
   Conference on Language Modeling (**COLM**), (**NENLP Outstanding Paper**), 2025
6. Automated Molecular Concept Generation and Labeling with Large Language Models
   Zimin Zhang*, Qianli Wu*, Botao Xia*, Fang Sun, Ziniu Hu, Yizhou Sun, **Shichang Zhang** (*equal contribution)
   International Conference on Computational Linguistics (**COLING**), 2025
7. An Explainable AI Approach using Graph Learning to Predict ICU Length of Stay
   Tianjian Guo, Indranil Bardhan, Ying Ding, **Shichang Zhang**
   Information Systems Research (**ISR**), 2024
8. Motif-driven Contrastive Learning of Graph Representations
   Shichang Zhang*, Ziniu Hu*, Arjun Subramonian, Yizhou Sun (*equal contribution)
   IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2024
9. Predicting and Interpreting Energy Barriers of Metallic Glasses with Graph Neural Networks
   Haoyu Li*, **Shichang Zhang***, Longwen Tang, Yizhou Sun (*equal contribution)
   International Conference on Machine Learning (**ICML**), 2024
10. PaGE-Link: Graph Neural Network Explanation for Heterogeneous Link Prediction
    Shichang Zhang, Jiani Zhang, Xiang Song, Soji Adeshina, Da Zheng, Christos Faloutsos, Yizhou Sun
    The Web Conference (**WWW**), 2023
11. GStarX: Explaining Graph Neural Networks with Structure-Aware Cooperative Games
    Shichang Zhang, Yozen Liu, Neil Shah, Yizhou Sun
    Advances in Neural Information Processing Systems (**NeurIPS**), 2022
12. Graph-less Neural Networks, Teach Old MLPs New Tricks via Distillation
    Shichang Zhang, Yozen Liu, Yizhou Sun, Neil Shah
    International Conference on Learning Representations (**ICLR**), 2022
13. Weak Models Can be Good Teachers: A Case Study on Link Prediction with MLPs
    Zongyue Qin, **Shichang Zhang**, Mingxuan Ju, Tong Zhao, Neil Shah, Yizhou Sun
    Learning on Graphs Conference (**LOG**), 2025
14. FUSE: Measure-Theoretic Compact Fuzzy Set Representation for Taxonomy Expansion
    Fred Xu, Song Jiang, Zijie Huang, Xiao Luo, **Shichang Zhang**, Yuanzhou Chen, Yizhou Sun
    Findings of the Association for Computational Linguistics (**ACL Findings**), 2024
15. SciBench Evaluating College-Level Scientific Problem-Solving Abilities of Large Language Models
    Xiaoxuan Wang*, Ziniu Hu*, Pan Lu*, Yanqiao Zhu*, Jieyu Zhang, Satyen Subramaniam, Arjun R Loomba, **Shichang Zhang**, Yizhou Sun, Wei Wang (*equal contribution)
    International Conference on Machine Learning (**ICML**), 2024

16. Laplacian Score Benefit Adaptive Filter Selection for Graph Neural Networks
Yewen Wang, **Shichang Zhang**, Junghoo Cho, Yizhou Sun
SIAM International Conference on Data Mining (**SDM**), 2024
17. Linkless Link Prediction via Relational Distillation
Zhichun Guo, William Shiao, **Shichang Zhang**, Yozen Liu, Nitesh Chawla, Neil Shah, Tong Zhao
International Conference on Machine Learning (**ICML**), 2023
18. Graph Condensation for Graph Neural Networks
Wei Jin, Lingxiao Zhao, **Shichang Zhang**, Yozen Liu, Jiliang Tang, Neil Shah.
International Conference on Learning Representations (**ICLR**), 2022

Preprints:

1. User Persona Subspaces Modulate Refusal Behavior in Language Models
Yan Zhou, **Shichang Zhang**, Zidi Xiong, Himabindu Lakkaraju
(MechInterp@ICML), 2026
2. How Faithful Is Trajectory-Based Data Attribution? Error Sources, Remedies, and Practical Guidelines
Junwei Deng, Pingbang Hu, Suliang Jin, Hao Lu, Jiachen T. Wang, **Shichang Zhang**, Jiaqi W. Ma
(Under Review), 2026
3. Towards Unified Attribution in Explainable AI, Data-Centric AI, and Mechanistic Interpretabil ity
Shichang Zhang, Tessa Han, Usha Bhalla, Himabindu Lakkaraju
(Under Review), 2025
4. Generalized Group Data Attribution
Dan Ley, **Shichang Zhang**, Suraj Srinivas, Gili Rusak, Himabindu Lakkaraju
(Under Review), 2024
5. Computational Copyright: Towards A Royalty Model for Music Generative AI
Junwei Deng, Xirui Jiang, Shiyuan Zhang, **Shichang Zhang**, Himabindu Lakkaraju, Ruijiang Gao, Chris Donahue, Jiaqi W. Ma
(Under Review), 2025
6. From Indirect Object Identification to Syllogisms: Exploring Binary Mechanisms in Transformer Circuits
Karim Saraipour, **Shichang Zhang**
(MechInterp@NeurIPS), 2025
7. Hierarchical Compression of Text-Rich Graphs via Large Language Models
Shichang Zhang, Da Zheng, Jiani Zhang, Qi Zhu, Xiang Song, Soji Adeshina, Christos Faloutsos, George Karypis, Yizhou Sun
(Preprint), 2024
8. Self-Control of LLM Behaviors by Compressing Suffix Gradient into Prefix Controller
Min Cai, Yuchen Zhang, **Shichang Zhang**, Fan Yin, Difan Zou, Yisong Yue, Ziniu Hu
(MechInterp@ICML), 2024
9. Parameter-Efficient Tuning Large Language Models for Graph Representation Learning
Qi Zhu, Da Zheng, Xiang Song, **Shichang Zhang**, Bowen Jin, Yizhou Sun, George Karypis
(Preprint), 2024

Media Coverage

Unifying AI Attribution: A New Frontier in Understanding Complex Systems — June 2025
HBS AI Insights

ChatGPT has entered the classroom: how LLMs could transform education — Nov. 2023
Nature News Feature

Teaching Experience

Instructor, University of California, Los Angeles
CS97: Introduction to Data Science — Summer 2024

Teaching Assistant, University of California, Los Angeles
CS145: Introduction to Data Mining — Fall 2020, Fall 2021
CS32: Introduction to Computer Science II — Spring 2021

Mentorship

Arjun Subramonian (UCLA Undergrad → UCLA PhD)	Mar. 2020 - Mar. 2021
Qianli Wu (UCLA Undergrad → Amazon SDE)	Mar. 2023 - Mar. 2024
Haoyu Li (UCLA Undergrad → UIUC PhD)	Mar. 2023 - July 2024
Gaotang Li (UMich Undergrad → UIUC PhD)	Oct. 2023 - June 2024
Botao Xia (UCLA Undergrad → UCLA Master)	Oct. 2023 - Aug. 2024
Zimin Zhang (UCLA Undergrad → UIUC Master)	Oct. 2023 - Dec. 2025
Min Cai (Shenzhen University Master → UAlberta PhD)	Nov. 2023 - Dec. 2025
Hongzhe Du (UCLA Master)	Mar. 2024 - Dec. 2025
Karim Saraipour (UCLA Master → Amazon SDE)	Apr. 2024 - Dec. 2025
Weikai Li (UCLA Ph.D.)	Sept. 2024 - Present
Dan Ley (Harvard Ph.D.)	Sept. 2024 - Dec. 2025
Ethan Ji (UCLA Master → UCLA PhD)	June 2025 - Dec. 2025
Terry Zhou (Harvard Master)	Sept. 2025 - Present
Hugo Tierrablanca (Harvard Master)	Feb. 2026 - Present
Yihan Wang (THU Undergrad)	Mar. 2026 - Present

Talks

A Holistic Scientific Understanding for Trustworthy AI — Jan - Feb 2026
Arizona State University
Mohamed bin Zayed University of Artificial Intelligence
University of Tennessee Knoxville
Pennsylvania State University
University of Massachusetts Boston

Explain AI Models: Methods and Opportunities in Explainable AI, Data-Centric AI, and Mechanistic Interpretability
NeurIPS Tutorial — Dec 2025

How Post-Training Reshapes LLMs
New England NLP Meeting — Apr 2025

Peering into The Mind of AI
Seminar at Georgia Institute of Technology — Apr 2025

Interpreting AI Systems Through Features, Data, and Model Components
Data Mining Seminar at Emory University — Apr 2025

Explainable AI for Graph Data and More
AI4LIFE Group at Harvard — Feb 2024

Graph Neural Network Explanation for Heterogeneous Link Prediction
Amazon Trans.AI Research Talks — July 2023
International World Wide Web Conference — May 2023

Structure-Aware Graph Neural Network Explanation
AI Time NeurIPS Talk Series — Feb 2023

Graph-less Neural Networks
NVIDIA GNN Reading Group — May 2022

Academic Service

Conference Area Chair:
NeurIPS - Advances in Neural Information Processing Systems — 2026
ACL ARR - Association for Computational Linguistics Rolling Review — 2025

Conference Reviewer/Program Committee:
NeurIPS - Advances in Neural Information Processing Systems — 2021 - 2025
ICML - International Conference on Machine Learning — 2022 - 2025
ICLR - International Conference on Learning Representations — 2024 - 2026
KDD - ACM SIGKDD Knowledge Discovery and Data Mining — 2020, 2023 - 2026
AAAI - AAAI Conference on Artificial Intelligence — 2023 - 2025
WSDM - ACM International Web Search and Data Mining Conference — 2023 - 2025
CIKM - ACM Conference on Information and Knowledge Management — 2022 - 2023
COLM - Conference on Language Modeling — 2026
ACL ARR - Association for Computational Linguistics Rolling Review — 2026
SDM - SIAM International Conference on Data Mining — 2024
LOG - Learning on Graphs Conference — 2023
ICDM - IEEE International Conference on Data Mining — 2021

Journal Reviewer:
Management Science
TPAMI - IEEE Transactions on Pattern Analysis and Machine Intelligence
TKDD - ACM Transactions on Knowledge Discovery from Data
TKDE - IEEE Transactions on Knowledge and Data Engineering
TNNLS - IEEE Transactions on Neural Networks and Learning Systems
TAI - IEEE Transactions on Artificial Intelligence

Workshop Organizer:
Workshop on Regulatable Machine Learning @ NeurIPS — 2024 - 2025

Reading Group Organizer:
UCLA Data Mining Reading Group — 2022 - 2024

Industry Work Experience

Amazon Web Service (AWS) — Santa Clara, CA
Applied Scientist Intern, Graph Machine Learning Team — June 2023 - Nov. 2023

- Proposed a framework for applying LLMs to text-rich graph data with hierarchical neighborhood compression, which allows LLMs to leverage the graph structure and handle long input text features gathered in a rich neighborhood.

- The proposed method outperformed traditional graph ML models on node classification benchmarks and will be incorporated into the Amazon DGL project.

Amazon Web Service (AWS) Santa Clara, CA
Applied Scientist Intern, Graph Machine Learning Team June 2022 - Oct. 2022

- Proposed a new framework to explain GNN link prediction for recommendation on graph data, which improves user trust in the model and helps developers debug the model. Work published in WWW 2023.
- The implemented framework will be incorporated into the Amazon Neptune ML project in production.

Snap Research Los Angeles, CA
Research Intern, Computational Social Science Team June 2021 - Sept. 2021

- Proposed a cross-model distillation framework to transfer knowledge from GNNs to MLPs, which speeds up model inference by 179 times and facilitates model deployment on latency-constraint applications. Work published in ICLR 2022.
- Worked on condensing large-scale training graphs to small synthetic graphs by over 90% reduction rate while maintaining competitive model performance for GNNs trained from scratch, which significantly saves storage space and achieves efficient continual learning. Work published in ICLR 2022.

WeWork Inc. Palo Alto, CA
Data Scientist Intern, Research and Applied Science Team June 2019 - Sept. 2019

- Implemented a data processing pipeline in SQL and Python for data querying, data cleaning, and feature engineering.
- Trained a Gradient Boosted Tree model on two million customer data to predict occupancy rate for WeWork buildings and achieved 0.093 MAE on the test set.
- Presented the pricing model as a selected outstanding project to the Research and Applied Science team including the VP.

Skills

Programming: Python (PyTorch, Hugging Face, DGL), C++, R, Java, Linux, Git
Natural Languages: Mandarin Chinese (Native), English (Proficient)